Atlas Explorer

A Research Tool for Atlas Cartography

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Atlas Explorer Project

The **aim** of the Atlas Explorer project is to

Identify

Analyze

Classify

Archive

Electronic Atlases (EAs) *

* An umbrella term encompassing all types of atlases on digital devices

STAGE 1

Interactive Web Catalog

Why Use Atlas Explorer?

Atlas Cartographers

Tracking trends in EA development

Analyzing EAs using different approaches



Preserving, **systematizing**, and **expanding knowledge** about EAs

Advancing atlas theory



Improving the quality of new EAs

General Public

Finding EAs

Educational purposes:

- What is an EA?
- What are the components of EAs?
- How to use EAs?
- How to critically interpret EA content?

What is an Atlas?

Collection of up-to-date **datasets** ← Earth sciences • GIS

Collection of **maps** (and related multimedia) ← Cartography

Visualization environment or Collection of visualizations ← Data Viz • Geovisualization

Narrative or Collection of stories ← Humanities • Critical cartography

Encyclopedia or Collection of articles ← Non-spatial sciences

System or **Model** ← Post-Soviet cartography • Geocybernetics

Website or **Application** ← **Atlas Explorer**

Atlas Definitions

Conceptual definition

Website or application that contains interrelated and coherent visual and geospatially referenced multimedia content organized in a logical structure within a specific topic and spatiotemporal frame to create an environment for (1) knowledge acquisition, (2) making comparisons and establishing relationships, (3) presenting research findings, (4) supporting decision-making, or (5) storytelling

Operational definition

Website or application that structures and/or unfolds a specific theme with a focus on spatiotemporal aspects

Atlas Criteria

- ✓ **Defined** theme and space
- ✓ **Includes** multiple thematic sections*
- ✓ **Includes** purpose-built maps or (geo)spatial visualizations
- ✓ **Content** is delivered either within the atlas shell or on purpose-built pages (not PDF pages)
- ✓ **Operational** and completed product (not a beta version)
- ⃠ **Not** a catalog geoportal, collection of dashboards, interactive map, (web)GIS or applied CIS

Database Status	
Identified	291
Added	100
Excluded	46
Pending	145

* *excluding the metaconcept of organizational mechanism*

Approaches to Atlas Analysis

C O N T E X T

1 Critical approach

CONTENT

2 Thematic

5 Thematic-semiotic

6 Thematic-cartographic

6.1 Geographic-cartographic (Geosystemic)

REPRESENTATION

3 Cartographic

4 Semiotic / Linguistic

7 Media

7.1 **Media-cartographic**

8 Narrative

8.1 Narrative-cartographic

TECHNICAL IMPLEMENTATION

9 Software-functional

10 System-architectural

Media-cartographic Analysis

Analysis Dimensions

Interface	**Information Architecture and Navigation**
Content Representation	**Functionality**

Method Quantitative Content Analysis (QCA) + Open-ended Questions

Purpose

- Determine the implementation of the EA components
- Define the metaconcept of EA
- Compare the EA concept with its practical implementation
- Compare EAs

Codes

755 binary
89 three-state
51 categorical
10 ordinal
117 open-ended

Duration

30-50 hours per EA

Research Questions

Bibliographic	*Which institutions and countries are the main publishers of EAs?* *How many EAs have been developed using atlas platforms?*
Content (Thematic)	*What are the most common topics and their combinations?* *What is the distribution of EAs by content spatiality?*
Media-cartographic	*What percentage of EAs include multiple classifications & filtering?* *What is the prevailing map concept in EAs?* *How many EAs support comparisons?*
Synthetic	*What are the distinctions between EAs of different generations?* *Does thematic focus influence the choice of EA metaconcept?*

Analysis Principles

Attribute Selection

- Comprehensive bibliographic coverage
- Representative content and media-cartographic attributes **describing the EA as a whole**

Sequential Workflow

- Predominantly **linear step-by-step** procedure
- Higher-level attributes derived from combinations of attributes

Reduced Subjectivity

Attributes focus on **implementation characteristics** rather than quality assessment

Express Analysis

Analysis time limited to **2–3 hours per EA** to maximize atlas coverage

Coding Scheme

43 binary
46 categorical
15 open-ended
4 three-state
10 ordinal

Attribute Definitions

Each attribute and value has an explicit definition available in the **glossary** and **table-header tooltips**

Bibliographic Attributes

Codes	
Binary	3
Three-state	0
Categorical	10
Ordinal	0
Open-ended	12

Updates ⟷ Major Updates

Publisher Type ⟷ Type of App Developer

Project Type ⟷ Paper Version

Distrubution ⟷ Access

Technology Type

Content Attributes

Thematic Coverage ↔ Topic 1, Topic 2, Topic 3 ↔ Theme

Codes	
Binary	5
Three-state	0
Categorical	8
Ordinal	2
Open-ended	2

Spatial Ontology of Objects & Phenomena + Geospatial Reference → Content Spatiality

- Cartographic
- Geospatial
- Spatial
- Graphical

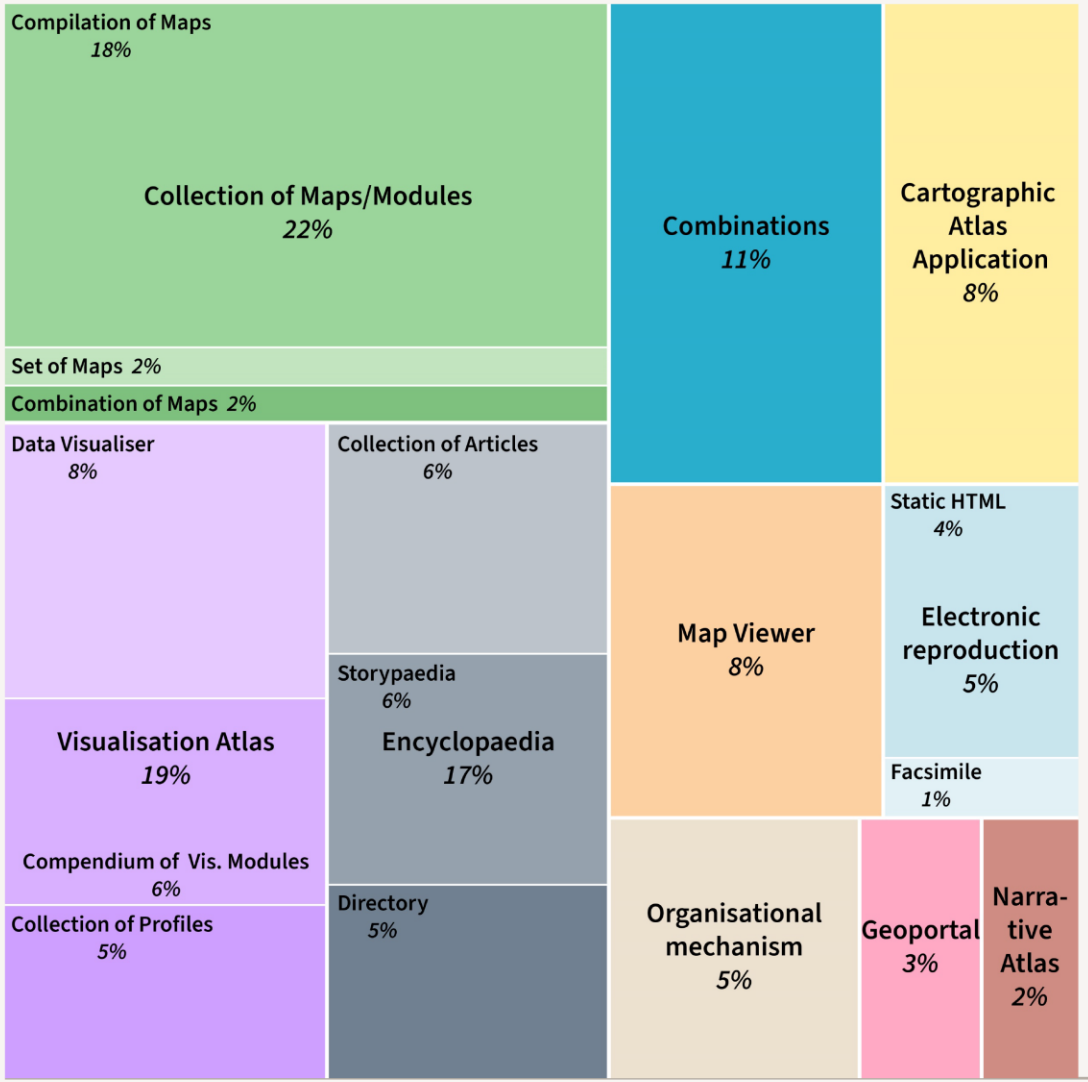
Spatial Coverage

AI-generated Content

Temporal Scope

Regular Updates

Media-Cartographic Attributes



Atlas Metaconcepts

Codes	
Binary	35
Three-state	4
Categorical	28
Ordinal	8
Open-ended	1

(1) Definition

(2) Description of EA components

(3) Typical usage scenario

(4) Target users and context of use

(5) Advantages and disadvantages

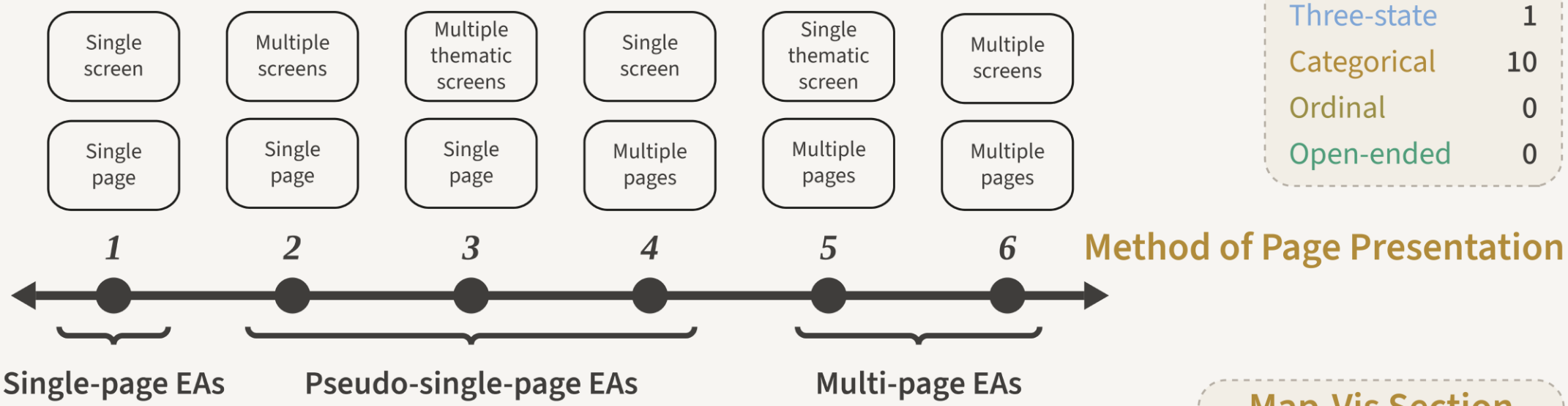
(6) Potential and areas for improvement

(7) Examples of EAs with illustrations

Krakovskyi, S., & Kurach, T. (2026). Revisiting the electronic atlas concept: from theoretical model to metaconcepts of atlas applications. *International Journal of Cartography,* 1–25. https://doi.org/10.1080/23729333.2026.2682461

General Attributes

Codes	
Binary	0
Three-state	1
Categorical	10
Ordinal	0
Open-ended	0



Atlas Focus

Data

Theme

Usage Style

Reading & Localization

Viewing & Comparison

Interaction & Analysis

Target Audience

General Public

GIS-literate Amateurs

Domain Specialists

Domain Specialists with GIS Skills

Map-Vis Section Implementation

Map-Vis Section

Several Map Sections

Non-Map Sections

Combinations

Interface Attributes

Codes	
Binary	1
Three-state	2
Categorical	2
Ordinal	3
Open-ended	0

Interface Responsiveness

Layout Template

Thematic Design

Layout Flexibility

Layout Pattern (Atlas Level) ⟷ Layout Pattern (Map-Vis Level)

Map Area Ratio (Atlas Level) ⟷ Map Area Ratio (Map-Vis Level)

Information Architecture & Navigation

Codes	
Binary	6
Three-state	0
Categorical	7
Ordinal	3
Open-ended	1

Homepage



Non-Map Sections



Map-Vis Sections



Atlas Level

Map-Vis Level

Content Hierarchy (Atlas Level) ⟷ Content Hierarchy (Map-Vis Level)

Organizational Structure (Atlas Level) ⟷ Organizational Structure (Map-Vis Level)

Navigation Mechanism (Atlas Level) ⟷ Navigation Mechanism (Map-Vis Level)

Main Information Unit

Content Classifications

Subjective Organization Scheme

Inter-content Navigation

Content Represenatation & Functionality

Main Representation Type
Articles
Maps/Map Modules
Infographics
Interactive Visualizations
Profiles
Multimedia Collections

Main Style of Maps
Traditional
Laconic
Expressive
Visualizational
Antique
Artistic

Map Concept
Restricted
Restricted Flexible
Flexible
Hybrid

CR Codes	
Binary	21
Three-state	1
Categorical	8
Ordinal	0
Open-ended	0

FN Codes	
Binary	7
Three-state	0
Categorical	1
Ordinal	2
Open-ended	0

Scroll-based Representation

Map as Interface

Map Labels

Storytelling Techniques

Multivariate Maps

Map Comparison

Main Basemap Content

Fully Functional

Map-Vis Interactivity

Content Export

Archiving Atlases

1. Screenshots

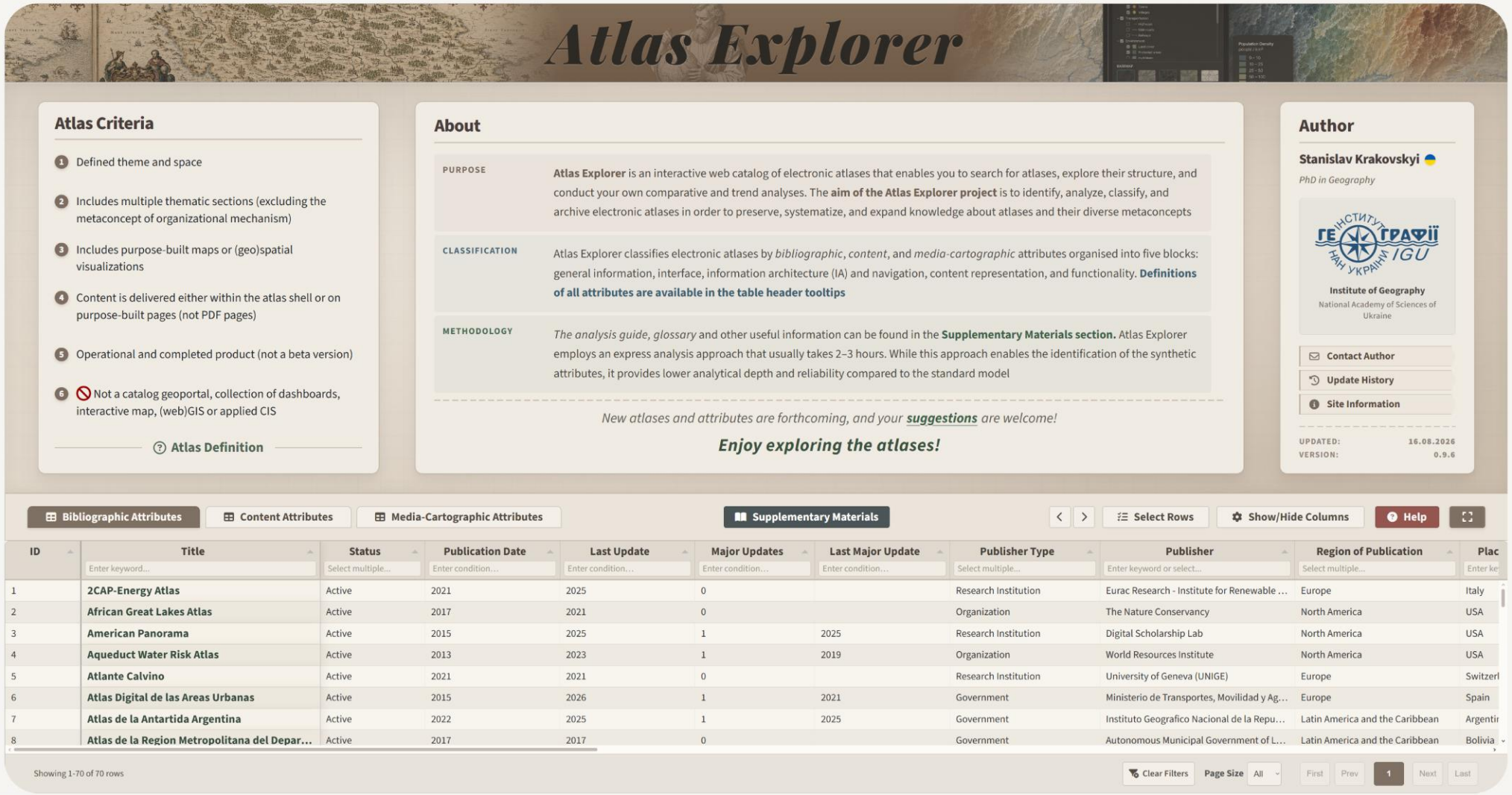


Homepage + Navigation pages + Content (non-map) pages + Map-Vis pages + Interesting solutions

2. Overviews of major trends (every 3-5 years)

3. Other options?...

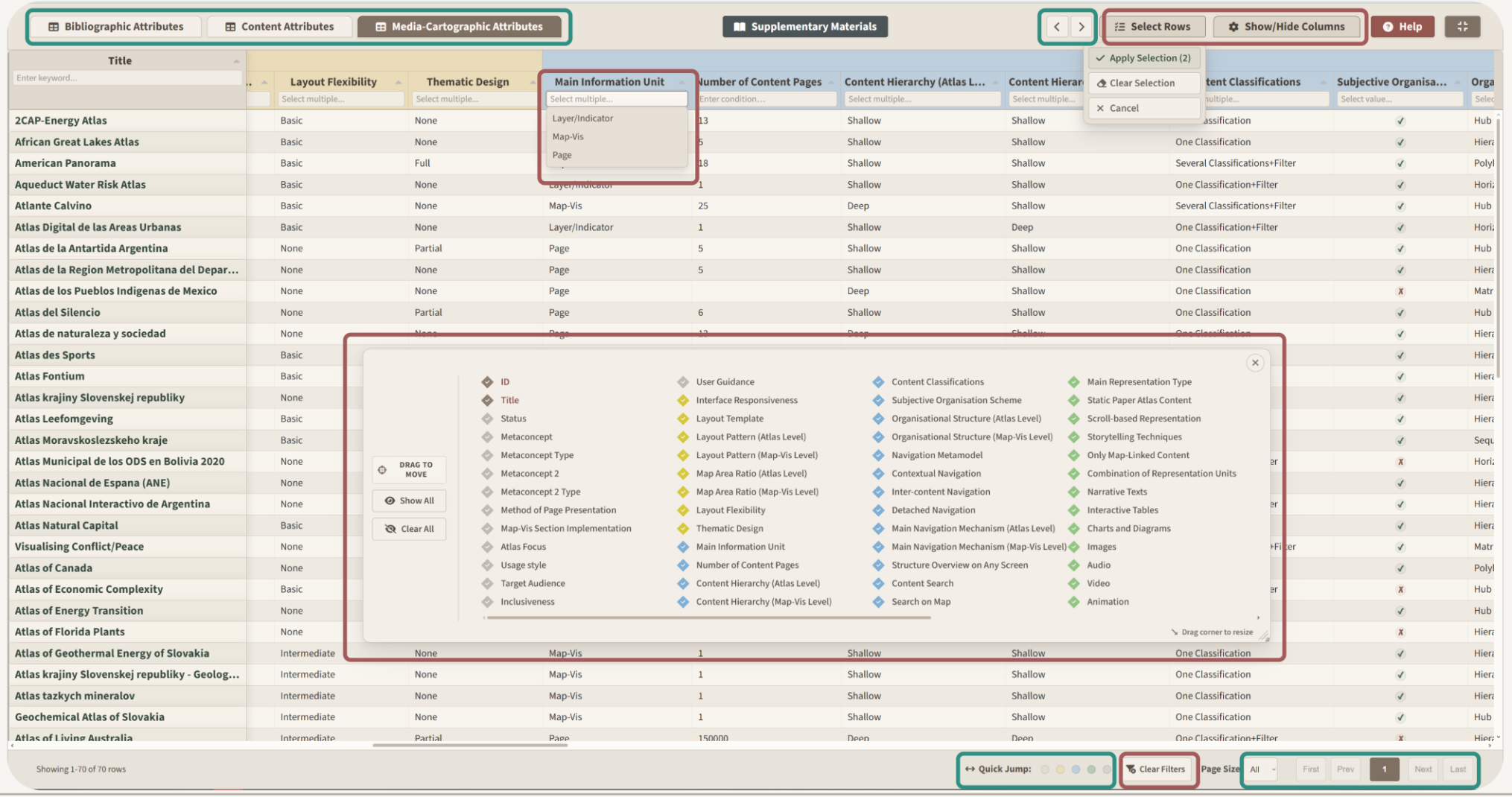
Atlas Explorer Website



Atlas Explorer Table Mode

Navigation

Filtering & Selection



Further Development

Stage I — Expansion

- Database expansion
- Screenshot gallery
- Non-atlas catalog
- Methodological resources

Stage II — MetaAtlas

- Map section
- Visualization section
- Enhanced interactive and analytical capabilities
- User participation in analysis

Thank you for your attention

Discover Atlas Explorer and get in touch

atlasexplorer.org

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